Dewi S. W. Gould

@dewi.s.w.gould@gmail.com, **List of Publications**

I am a **research lead** at Arcadia Alignment. I am currently studying automated alignment research (failure modes, monitoring, speed-up compared to R&D). I have worked on empirical AI debate, no-CoT time horizons, explanatory faithfulness, inverse reinforcement learning. I have a background in mathematics and theoretical physics: I obtained a PhD in string theory in 2024 from the University of Oxford.

SELECTED RESEARCH

Think Fast: Estimating No-CoT Task-Completion Time Horizons of Frontier AI Models
***Dewi S.W. Gould** et. al.*

A Positive Case for Faithfulness: LLM Self-Explanations Help Predict Model Behavior
*H Mayne, J.S. Kang, **Dewi S.W. Gould**, et. al.*
- Presented at **ICML**, 2026

PAC Apprenticeship Learning with Bayesian Active Inverse Reinforcement Learning
*O Bajgar, **Dewi S.W. Gould**, et. al.*
- Presented at **Reinforcement Learning Conference** 2025
- Presented at **NeurIPS Workshop** on Bayesian Decision Making, 2024

SKATE, a Scalable Tournament Eval: Weaker LLMs differential between stronger ones using verifiable challenges
***Dewi S.W. Gould**, Bruno Mlodozeniec, Samuel Brown*

BLOG POSTS

Most of my **recent work at Arcadia is on LW**.

EDUCATION

PHD in Mathematics

"Generalised Symmetries in String Theory Realizations of Quantum Field Theories"
University of Oxford (2020-2024)
- Research **summary**

MAst Applied Mathematics

University of Cambridge (2019-2020)
- *No numerical grades awarded due to COVID

MSci Theoretical Physics

Imperial College London (2015-2019)
- Highest rank final year student (97% final exam average)
- CMS Prize

SKILLS

Python, git, PyTorch.

EXPERIENCE

ASTRA AI SAFETY FELLOWSHIP, Constellation — January '26-April, Berkeley, USA.

Worked with **Redwood Research** on diffuse control and capability elicitation. Mentors: Julian Stastny and Alex Mallen.

SPAR AI SAFETY FELLOWSHIPS — January '25-January '26, Glasgow, UK.

Two SPAR AI safety projects. The first, on scalable LLM evaluations mentored by Bruno Mlodozeniec (Uni. Cambridge) and Sam Brown (Independent), and the second on explanatory faithfulness mentored by Noah Siegel (GDM).

POSTDOCTORAL RESEARCHER, Alan Turing Institute. — June '24-December '25, London, UK.

Led research on autonomous methods for creating novel and controllable air traffic scenarios using Large Language Models. Helped engineering efforts in building a digital twin of UK airspace. Studied complexity prediction methods using graph neural networks. See outreach **talk.** Led by Prof. Richard Everson.